

Helix Thready — Motion Package (Lottie catalogue)

Inventory

Reduced-motion fallbacks (static SVGs)

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Rev	Date	Author	Change
1	2026-07-22	swarm (design/motion)	Initial package: six hand-authored Lottie assets, motion spec, self-contained preview, manifest, validation record
2	2026-07-22	swarm (design · motion)	Static <code>reducedMotionFallback</code> SVGs delivered (asset half of <code>THREADY-MOT-03</code> closed — see <code>motion.md</code> §8 for the narrowed remainder); manifest v2 names them per animation; preview gains a static-fallback / reduced-motion demonstration toggle

This directory is the concrete **motion deliverable** of the design area — the “lots of Lottie animations” the request mandates (`prototypes.md` §6), realized as **hand-authored vector Lottie** (v5.x schema, shape layers only, no rasters, no design-tool export involved). It partially delivers the Lottie half of `[OPEN: THREADY-DES-02]`; see `motion.md` §8 for what remains open.

Upstream: `design-system.md` §5 (motion tokens, reduced-motion rule), `prototypes.md` §6 (choreography spec), `ux-flows.md` (the journeys each asset binds to). **Downstream:** app motion runtimes (`lottie-web` / `lottie-compose` / `lottie-ios`), the visual-regression bank (`[GAP: 9.3]`).

Inventory

File	What it is	Frames @ fps	Loop	Plays on	Validation
<code>helix-spinner.json</code>	The double-helix mark weaving/rotating (two phase-animated sine strands, lime + teal) — loading	120 @ 60 (2 s)	yes, seam-exact	app boot, route loads, resolve waits	parse · structure · lottie-web render
<code>success-check.json</code>	Lime disc pop + ink check drawn by trim-path + accent ring ping	90 @ 60 (1.5 s)	one-shot	<code>201</code> created, <code>200</code> saved, <code>processing.com</code> mpleted	parse · structure · lottie-web render
<code>error-cross.json</code>	Danger disc pop + snow cross drawn + shake + ring ping (<code>--danger</code> , never brand)	90 @ 60 (1.5 s)	one-shot	<code>processing.failed</code> , <code>422</code> / <code>403</code> , AA-fail	parse · structure · lottie-web render
<code>processing-pulse.json</code>	Breathing lime core + two staggered teal ripple rings — indeterminate work	120 @ 60 (2 s)	yes, seam-exact	pipeline step running/queued, reprocess in flight	parse · structure · lottie-web render
<code>thread-sync.js</code>	Messenger node ⇌ Thready node with dots shuttling along the thread — messenger sync	120 @ 60 (2 s)	yes, seam-exact	channel backfill / live sync, session established	parse · structure · lottie-web render
<code>transition-fade-slide.json</code>	Shared-axis X fade+slide between two cards at real 300 ms timing, with segment markers	90 @ 60 (1.5 s)	one-shot	route/page + wizard-step transitions (reference rendering)	parse · structure · lottie-web render
<code>spiral-static.svg</code>	Reduced-motion fallback for <code>helix-spinner</code> — the exact frame-15 poster state (interpolated strand paths + eased strand opacities from the Lottie keyframes)	static	—	<code>prefers-reduced-motion</code> swap	xmllint · geometry derived from Lottie keyframes
<code>check.svg</code>	Reduced-motion fallback for <code>success-check</code> — final frame 89 (disc at rest, check fully)	static	—	<code>prefers-reduced-motion</code> swap	xmllint · geometry derived from Lottie keyframes

File	What it is	Frames @ fps	Loop	Plays on	Validation
	drawn; invisible ring ping omitted)				
<code>cross-static.svg</code>	Reduced-motion fallback for <code>error-cross</code> — final frame 89 (disc at rest after shake, cross fully drawn; invisible ring ping omitted)	static	—	<code>prefers-reduced-motion</code> swap	xmllint · geometry derived from Lottie keyframes
<code>pulse-static.svg</code>	Reduced-motion fallback for <code>processing-pulse</code> — the exact frame-30 poster state (core at 114 %, both rings at their interpolated scale/opacity)	static	—	<code>prefers-reduced-motion</code> swap	xmllint · geometry derived from Lottie keyframes
<code>sync-static.svg</code>	Reduced-motion fallback for <code>thread-sync</code> — the exact frame-30 poster state (nodes + thread, carrier dot at its eased x 365.78)	static	—	<code>prefers-reduced-motion</code> swap	xmllint · geometry derived from Lottie keyframes
<code>motion.md</code>	The motion spec: tokens, interaction map, reduced-motion contract, runtime integration, honest validation record	—	—	—	—
<code>motion-manifest.json</code>	Runtime manifest v2 (ids, loop, poster frames, reduced-motion mode + per-animation <code>reducedMotionFallback</code> SVG, segment markers) — concrete form of the prototypes §6 excerpt	—	—	—	parse
<code>preview.html</code>	Self-contained preview: vendored lottie-web 5.13.0	—	—	—	jsdom smoke test (re-run after fallback toggle) · Chromium

File	What it is	Frames @ fps	Loop	Plays on	Validation
	(MIT) inline + the six JSONs embedded verbatim; theme toggle, per-asset scrub/replay, reduced-motion poster behavior, and a “Show static fallbacks” toggle (auto-engaged under <code>prefers-reduced-motion</code>) with the five fallback SVGs embedded inline. No CDN, zero network requests.				light/dark/reduced-motion screenshots (rev 1 page)
<code>diagrams/motion-asset-map.mm.d</code>	Mermaid source of the journey→asset map in motion.md §4	—	—	—	—

Validation meaning (full record: motion.md §7): `parse` = `python3 json.load`; `structure` = required top-level keys (`viewport` layers), 5.x schema, `ty: 4` shape layers only with non-empty `shapes`, monotonic keyframes, colors in [0, 1], empty `assets`, exact loop seams; *lottie-web render* = loaded and frame-seeked in lottie-web 5.13.0 (SVG renderer, headless under jsdom) producing real geometry, with animation-progression assertions (trim draws, transforms move, shapes morph). All six were additionally rendered in **headless Chromium** (real Blink/Skia compositor) as a multi-frame light+dark contact sheet and reviewed. **Not** verified: physical-display pixels, `lottie-compose` / `lottie-ios` runtimes — tracked `[OPEN: THREADY-MOT-01]`.

Reduced-motion fallbacks (static SVGs)

Each playable animation names its static fallback in `motion-manifest.json` (`reducedMotionFallback`); the SVG renders the **exact state the reduced-motion contract freezes on** — the poster frame for the loopers, the final frame for the one-shots — with geometry taken from the Lottie keyframes (interpolated where the frame falls between keyframes; see each SVG’s header comment for the derivation). `transition-fade-slide` deliberately has `reducedMotionFallback: null` — its reduced-motion behavior is an `instant-cut`, no artwork exists on purpose. The `empty-channels.svg` fallback named in the prototypes §6 excerpt belongs to the illustration-grade set that is still open under `[OPEN: THREADY-DES-02]` and is therefore intentionally absent.

Coloring `[DEFAULT — adjustable]`: fills/strokes are written as `style="fill: var(--brand, #B6E376)"` — the literal fallback values are the Lottie’s own baked Thready colors (dark-surface tuned, [motion.md §6](#)), so a standalone `<img>` load matches the animations exactly, while inlining the SVG into a token-bearing page themes it via `--brand` / `--brand-2` / `--brand-ink` / `--danger` — the SVG analogue of the documented load-time remap. `currentColor` was not usable: these are multi-color assets. The snow cross strokes in `cross-static.svg` stay literal `#F8FAFC` by design (must remain light-on-danger in both themes; no on-danger token exists).

How to preview

Open `preview.html` in any browser — it is fully self-contained (player vendored inline; works offline). Or load any `.json` into a standard Lottie player. With `prefers-reduced-motion: reduce` active, the page swaps every slot to its static fallback SVG (the production behavior); the header’s “Show static fallbacks” / “Show animations” toggle demonstrates the swap in either mode, per the [motion.md §5](#) contract.

Color provenance

All colors are the Thready theme tokens ([design-system.md §3.2](#)) in Lottie 0–1 RGBA form:

Token	Hex	Lottie array	Used in
<code>--brand</code> (lime)	#B6E376	[0.714, 0.890, 0.463]	spinner strand, check disc, pulse core, sync node/dot, incoming card
<code>--accent</code> light	#446E12	[0.267, 0.431, 0.071]	success ring ping; light-theme remap target (motion.md §6)
<code>--brand-2</code> light (teal)	#ABDDC9	[0.671, 0.867, 0.788]	spinner strand, ripple ring A, sync node/thread, outgoing card
<code>--brand-2</code> dark (teal)	#B7EBD6	[0.718, 0.922, 0.839]	ripple ring B, sync return dot
<code>--brand-ink</code>	#0A0F04	[0.039, 0.059, 0.016]	check stroke on the lime disc (13.15:1 — the documented pairing, design-system §3.2)
ink (<code>--fg</code> light)	#020817	[0.008, 0.031, 0.090]	card text bars (at 30 % fill opacity)
snow (<code>--fg</code> dark)	#F8FAFC	[0.973, 0.980, 0.988]	cross strokes
<code>--danger</code> light	#DC2626	[0.863, 0.149, 0.149]	error disc + ping — semantic error is never brand-colored (design-system §6)

The set is tuned for dark surfaces (lime **is** the dark accent); light-surface and white-label retinting is a documented load-time remap, [motion.md §6](#).

How these were authored

Hand-authored as Lottie JSON (no After Effects, no Figma plugin, no downloads) via a deterministic generator script that computes the geometry — sine-through-bezier strand keyframes with analytic tangents for the spinner, visibility-aware seam-exact keyframes for the loopers — and every eased keyframe carries the shipped token curve `--ease-standard cubic-bezier(0.2, 0, 0, 1)` [[VERIFIED — design-system §5](#)]. The validator that gated them (and caught two real seam defects during authoring) is part of the record in [motion.md §7](#).

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